

Study, career, and some science

Notes from talk of Prof.V. Balakrishnan

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1 Career guidelines

- We should have goals in life and we should be able to work towards our goals to achieve them.
- If we have confusion in making career decisions, then be happy as it is a sign of intelligence. We should be able to explore all options, as in present times since our longevity is high (80-90 years), so we don't have to worry about retirement plans, and we can pursue 2-3 careers in one lifetime. We can work when feel like working, there won't be any standard working times too in the future.
- Young people should not settle down very fast, you should make some wrong turns, backtrack a little and continue. Remember only sediments settle down fast and everything else is in flux and that is the way life should be, People should be able to dream and they should be able to achieve it.
- The sky is not the limit, but remember nothing comes easy, every single thing you can achieve comes very hard, and there is no substitute for hard work. You should think about your work most of the time and look for opportunities. There is a lot of pain associated with doing hard and creative and, you should have a little bit of discontent with yourselves that you can do better. Only potatoes and vegetables are completely happy with their state of life. That discontent should be directed towards ourselves and we should not compare with other human beings. There is no point in comparison with other people, Your competitor is 'YOU' and nobody else.

- Don't be elated at the smallest achievement and stop working altogether, nor should your small failures affect you to the stage that you become completely depressed.
- Those who do best in the future are not necessarily toppers of exams who write notes and remember those notes and succeed. These are those people who go from one student to another, clearing doubts, having discussions in-group, such people succeed in the end and that's what we should aspire to be.

2 On Science and understanding :

- All knowledge is not in books, most questions haven't been answered. Science is the attempt to know the whole universe by asking questions and trying to figure out the solution.
- Science has no authority figures, as nobody has the right to tell nature how to behave. Science is self-correcting and is subjected to scrutiny all the time.
- Homeopathy and astrology are bogus and pseudoscience. Everything is connected but the claim of astrologers to predict the future in highly chaotic world is bogus.
- The universe seems to be mathematically constructed. As math is the language and logic built in it and it is exactly what we need to deal with the universe.
- We should ask as many questions. There is no such thing as stupid questions. We Should not hesitate to expose our ignorance as long as we want to know. A lot of progress in science comes from our courage to ask new questions. If you don't know something, ask that to a knowledgeable person, and got it clarified, because a state of ignorance is uncomfortable one, because we know under the pillow that we don't know the answer.
- Our understanding is unique, no two people understand something in the same way. All information is processed in all different brains slightly

differently. Some people think geometrically in form of pictures, some see things algebraically, they think in symbols, equations. Some people think analogically and metaphorically.

- Today science is more or less an open book that requires a great deal of discussion, collaboration, and above all teaching people. Knowledge is the only thing that grows when you share it. To teach other people, your peers, children, younger siblings. Every time you teach, your knowledge improves and your understanding of 'your knowledge' improves. Remember real understanding comes from the teaching, absorb enough knowledge and emit equal in an amount too.
- Exams are necessary evils and it is not the best way to check the knowledge and understanding of students, sadly we are just stuck with this system. The exam system has one positive effect that it disciplines us by having deadlines. “ Exams does not check your knowledge/understanding but discipline, If we have been disciplined throughout the year, we could have gained enough what is required to pass the exams with good discipline scores.”